Hierarchical connectivity sets the upper critical dimension of neural avalanches

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Neural avalanches with size exponent 3/2 are usually explained by mean-field critical branching, which for short-range coupling requires dimension $N \geq 4$. We show that hierarchical connectivity removes this constraint. In a p-adic version of the Buice–Cowan field theory, a coupling that decays by $p^{-(N+\sigma)}$ per hierarchical level selects the Taibleson–Vladimirov operator, and the critical theory is ultrametric directed percolation with upper critical dimension $N_c = 2\sigma$. Exact simulations on a tree give $\tau_S = 3/2$ for $\sigma \leq N/2$, decreasing continuously for $\sigma > N/2$.

Cortical activity propagates in cascades, neuronal avalanches, whose sizes are distributed as $S^{-3/2}$ [1]. The standard explanation is that the cortex operates near the critical point of an absorbing-state transition, and that 3/2 is the mean-field avalanche exponent of a critical branching process. Buice and Cowan made this precise: their field theory of spiking networks maps onto Reggeon field theory, so the transition is in the directed-percolation (DP) class [2]. This raises a difficulty. For short-range coupling, DP has upper critical dimension $N_c = 4$, and below it the avalanche exponent differs from 3/2; in one dimension, $\tau_S \simeq 1.11$ [3]. A cortical sheet is effectively two-dimensional (three at most, counting layers), well below $N_c = 4$; here, as in Ref. [2], N counts the dimensions of the neuron space, and time is an additional coordinate. In a Euclidean theory with local coupling, cortical avalanches would therefore not be mean-field. But cortical connectivity is hierarchical. Sporns and co-workers proposed that the brain's structural connections are arranged in self-similar, nested patterns across multiple spatial scales, from local circuits to whole brain regions [4–6]. Such an architecture keeps the physical wiring cost low while supporting high functional complexity and information-processing capacity [7, 8]. This fractal-like self-similarity has no natural counterpart in $\mathbb{R}^N$ with local coupling, but it is built into $\mathbb{Q}_p^N$, where $\mathbb{Q}_p$ denotes the field of p-adic numbers, which naturally form a tree-like, self-similar network. Here we show that this hierarchy, not the dimension of space, decides whether avalanches are mean-field. We formulate the Buice–Cowan theory on a hierarchical neuron space, the p-adic numbers, and find that the upper critical dimension is $N_c = 2\sigma$, where σ measures how fast the coupling decays up the hierarchy. Any $\sigma > 0$ is allowed, so the 3/2 law holds even for $N = 1$ when $\sigma \leq 1/2$, and exact simulations confirm the predicted crossover. The mathematical details, including the exact discretization, the operator-selection and uniqueness results, the heat-kernel estimates, the derivation of the Ginzburg criterion and the one-loop equations, are worked out in a companion paper [9], which also contains the further results mentioned below.

p-Adic neuron spaces. Ultrametric spaces entered physics through the theory of spin glasses: in Parisi's solution of mean-field models, the pure states are organized in an ultrametric hierarchy [10]. The p-adic numbers are the prototypical ultrametric spaces on which one can do analysis, and they have been used to formulate replica symmetry breaking [11] and relaxation in hierarchical energy landscapes [12]. Here the hierarchy is not in the space of states but in the architecture: the neurons themselves are the points of an ultrametric space. Fix a prime p. Every nonzero p-adic number has a unique expansion $x = p^{\gamma}\sum_{j\geq 0} x_j p^j$, with digits $x_j \in \{0, \ldots, p-1\}$, $x_0 \neq 0$, and $|x|_p = p^{-\gamma}$ [13, 14]. The norm satisfies the strong triangle inequality $|x+y|_p \leq \max\{|x|_p, |y|_p\}$. Consequently two balls are either disjoint or nested, and the unit ball $\mathbb{Z}_p$ is the boundary of a rooted tree in which each vertex has p children. The distance $|x-y|_p$ records the level of the last common ancestor of x and y (Fig. 1). In dimension N we use $\mathbb{Q}_p^N$ with $\|x\|_p = \max_i |x_i|_p$; it carries a Haar measure $d^N x$, a Fourier transform and a theory of pseudodifferential operators [13, 14]. A finite network of depth l has as neurons the p^{Nl} elements of $G_l^N = (\mathbb{Z}_p/p^l\mathbb{Z}_p)^N$, the leaves of a finite tree of depth l; the internal vertices are modules (balls). We say that the neurons are organized in a tree-like structure, but a hierarchical network includes a tree as a particular case. In general, every pair of neurons is coupled, with a strength $w_{ij} = w(\|i-j\|_p)$ fixed by the level of their last common ancestor [Fig. 2(a)], and the neuron space has finite dimension N, since a ball of radius p^k has Haar volume p^{kN}.

Model. We use the effective spike model of Buice and Cowan [2]: each effective spike decays at rate α, and neuron i emits spikes at rate $f\big(\sum_j w_{ij} n_j\big)$, with $f(0) = 0$ (an absorbing state), $f'(0) > 0$ and $f''(0) < 0$ (saturation). The absorbing state is an idealization: resting cortex receives weak input, $f(I) > 0$, and is described by a weakly driven state near criticality, in which input spikes start avalanches; the absorbing state is its limit of

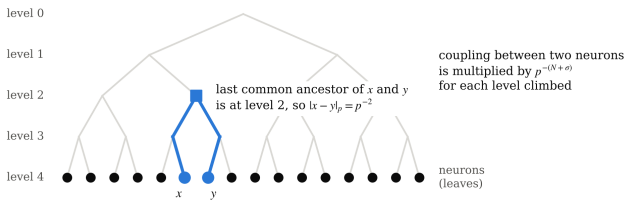


FIG. 1. Neurons organized in a tree-like structure ($p = 2$, depth 4: levels $0, \dots, 4$). The p-adic distance between two neurons is determined by the level of their last common ancestor. With a kernel $w(\|x\|_p) \propto \|x\|_p^{-N-\sigma}$, the exponent σ measures how fast coupling decays with hierarchical distance.

zero input. The Doi–Peliti construction gives the action

$$S[\varphi,\tilde\varphi] = \int_{\mathbb{Q}_p^N} d^N x \int_0^\infty dt \Big\{ \tilde\varphi\, \partial_t \varphi + \alpha \tilde\varphi \varphi - \tilde\varphi\, f\big(w \star (\tilde\varphi\varphi + \varphi)\big) \Big\}, \tag{1}$$

with $(w\star g)(x) = \int w(\|x-y\|_p) g(y)\, d^N y$; the initial-state term is omitted. Unlike a lattice approximation of $\mathbb{R}^N$, the network G_l^N is an *exact* restriction of the continuum theory: for fields constant on the balls of radius p^{-l}, the discrete action equals Eq. (1) for every l [9]. Setting $\tilde\varphi = 0$ in the equations of motion gives the hierarchical Wilson–Cowan equation $\partial_t a = -\alpha a + f(w \star a)$.

The connectivity selects the diffusion operator. Linearizing around the quiescent state, a mode of momentum $\xi \in \mathbb{Q}_p^N$ relaxes at the rate $\lambda(\xi) = \mu + A_w(\xi)$, exactly, where $\mu = \alpha - f'(0)\hat w(0)$ is the bare distance to criticality and

$$A_w(\xi) = f'(0)\big(\hat w(0) - \hat w(\xi)\big) \tag{2}$$

is the symbol of the nonlocal operator $\boldsymbol{W}\phi(x) = f'(0)\int [\phi(x) - \phi(x-y)]\, w(\|y\|_p)\, d^N y$. Thus the free part of the theory is $\tilde\varphi(\partial_t + \mu + \boldsymbol{W})\varphi$, with no approximation.

In $\mathbb{R}^N$ the same construction gives an operator whose long-wavelength limit is $-D\nabla^2$ whenever w has a finite second moment: all short-range kernels reduce to the Laplacian. Over $\mathbb{Q}_p^N$ there is no such reduction, because locally constant functions have no Taylor expansion. Each connectivity kernel selects its own heat equation $\partial_t u + \boldsymbol{W} u = 0$, and the infrared physics depends on how $A_w(\xi)$ behaves as $\|\xi\|_p \to 0$. The natural hierarchical kernel attenuates the coupling by the same factor at every level,

$$\frac{w(p^{j+1})}{w(p^j)} = p^{-(N+\sigma)}, \qquad \text{i.e.} \quad w(\|x\|_p) = C_w \|x\|_p^{-N-\sigma}, \tag{3}$$

for $\|x\|_p$ larger than some p^r. Then, exactly for $\|\xi\|_p \le p^{-r}$,

$$A_w(\xi) = \kappa_W \|\xi\|_p^\sigma, \qquad \kappa_W = f'(0)\, C_w\, \frac{1 - p^{-\sigma-N}}{p^\sigma - 1}, \tag{4}$$

which is the symbol of $\kappa_W \boldsymbol{D}^\sigma$, with $\boldsymbol{D}^\sigma$ the Taibleson–Vladimirov operator [13, 14]. Conversely, $\boldsymbol{D}^\sigma$ is the only isotropic pseudodifferential operator on $\mathbb{Q}_p^N$ that is homogeneous of degree σ [9]. Self-similar hierarchical connectivity therefore fixes the diffusion operator up to one constant. Its heat kernel has power-law tails in hierarchical distance, $Z_t(x) \asymp \kappa_W t\, [(\kappa_W t)^{1/\sigma} + \|x\|_p]^{-N-\sigma}$, with constants that depend only on p, N and σ [9, 15].

Ultrametric directed percolation and $N_c = 2\sigma$. Expanding f to second order, shifting the response field and rescaling as in Ref. [2], the relevant part of Eq. (1) becomes

$$S_{\text{eff}} = \int d^N x\, dt \Big\{ \tilde\psi\big(\partial_t + \mu + \boldsymbol{W}\big)\psi + \frac{g}{2}\big(\tilde\psi\psi^2 - \tilde\psi^2\psi\big) \Big\}, \tag{5}$$

with $g = [2f'(0)|f''(0)|\hat w(0)^3]^{1/2}$. This is Reggeon field theory with $-D\nabla^2$ replaced by $\boldsymbol{W}$, and it keeps the rapidity-reversal symmetry $\psi(x,t) \leftrightarrow -\tilde\psi(x,-t)$, since $\boldsymbol{W}$ is self-adjoint. Power counting with $A_w \simeq \kappa_W \|\xi\|_p^\sigma$ gives $z = \sigma$ at tree level, $[\psi] = [\tilde\psi] = N/2$, and

$$[g] = \sigma - \frac{N}{2}, \qquad N_c = 2\sigma. \tag{6}$$

Above N_c the mean-field exponents

$$\beta = 1, \quad \nu_\parallel = 1, \quad \nu_\perp = \frac{1}{\sigma}, \quad z = \sigma, \quad \tau_S = \frac{3}{2} \tag{7}$$

are exact; at $N = N_c$ they acquire logarithmic corrections. Quartic and higher vertices are irrelevant near N_c, and a correction $O(\|\xi\|_p^{\sigma+\delta})$ to the symbol does not change the power counting.

The comparison with Euclidean long-range DP is instructive. With Lévy-flight spreading, $w(x) \sim |x|^{-N-\sigma}$ in $\mathbb{R}^N$, one also finds $N_c = 2\sigma$, but only for $\sigma < 2$: for $\sigma \ge 2$ the second moment is finite, ∇^2 takes over, and $N_c = 4$ [16, 17]. Over $\mathbb{Q}_p^N$ no local operator exists, so *every* $\sigma > 0$ defines a distinct universality class, and $N_c = 2\sigma$ ranges over $(0, \infty)$. In particular, for $N = 1$ the mean-field avalanche law $\tau_S = 3/2$ holds whenever $\sigma \le 1/2$, while for $N = 2$ it holds whenever $\sigma \le 1$. The situation is analogous to Dyson's hierarchical model [18, 19], which also has a finite upper critical dimension set by the decay of the coupling. It is not that of DP on a Cayley tree, which is mean-field for any coupling because the number of sites grows exponentially with graph distance.

Ginzburg criterion. The one-loop correction to the inverse propagator is small compared with μ if

$$\frac{g^2}{4} I_2(\mu) \ll 1, \qquad I_2(\mu) = \int_{\|\xi\|_p \le p^\Lambda} \frac{d^N \xi}{[\mu + A_w(\xi)]^2}. \tag{8}$$

$I_2(0)$ is finite if and only if $N > 2\sigma$, consistent with Eq. (6). For $N < 2\sigma$ (see Ref. [9] for the proof), summing over the spheres $\|\xi\|_p = p^k$ gives, as $\mu \to 0^+$,

$$I_2(\mu) \simeq (1 - p^{-N})\, \kappa_W^{-N/\sigma}\, \mu^{N/\sigma - 2}\, \Phi(s), \tag{9}$$

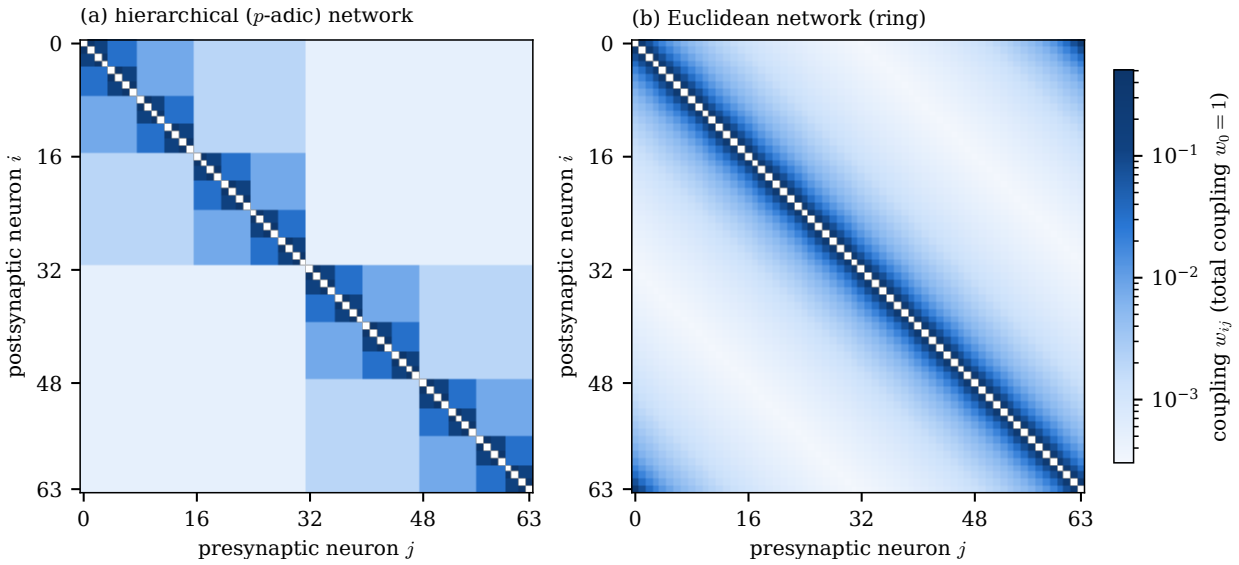


FIG. 2. Connection matrices w_{ij} of two networks of 64 neurons with $\sigma = 1$ (logarithmic color scale; diagonal excluded; total coupling per neuron normalized to 1). (a) Hierarchical network, $p = 2$, $N = 1$, $l = 6$, $w_{ij} \propto |i-j|_2^{-(1+\sigma)}$. Neurons are ordered along the boundary of the tree; the nested blocks are the modules of levels $5, 4, \ldots, 1$, and w is constant on each block. (b) Euclidean ring with $w_{ij} \propto |i-j|^{-(1+\sigma)}$.

where $\mu/\kappa_W = p^{\sigma s}$ and

$$\Phi(s) = \sum_{k\in\mathbb{Z}} \frac{p^{(k-s)N}}{\left(1+p^{\sigma(k-s)}\right)^2} \tag{10}$$

is periodic in s with period 1. With the ultrametric diffusion length $\ell_W = (\kappa_W/\mu)^{1/\sigma}$, the criterion reads

$$\frac{1-p^{-N}}{4}\,\frac{g^2}{\kappa_W^2}\,\ell_W^{2\sigma-N}\,\Phi(s) \ll 1. \tag{11}$$

The left-hand side is the relative size of the fluctuation correction to the response of the network, i.e., to the mean avalanche size. The coupling $g^2 \propto f'(0)|f''(0)|$ measures the spike-count noise that survives in the dynamics of the mean; it comes from branching and saturation, and vanishes for a linear f, for which mean-field theory is exact. The factor $\ell_W^{2\sigma-N}$ measures how much of this noise is averaged out: a perturbation lives a time $1/\mu = \ell_W^\sigma/\kappa_W$ and spreads over a ball of radius ℓ_W, which contains a number of neurons proportional to ℓ_W^N. As $\mu \to 0$, this ball climbs the hierarchy. If $N > 2\sigma$, the number of neurons it contains grows faster than the noise accumulates, and mean-field theory improves; if $N < 2\sigma$, the coupling decays too fast up the hierarchy, the effective number of independent inputs does not keep up, and fluctuations dominate. Three consequences follow. First, on the infinite network and for $N < 2\sigma$, mean-field theory fails sufficiently close to criticality, whatever the value of g. Second, on a finite network of diameter L, the left-hand side is largest at $\ell_W \approx L$, so mean-field theory fails inside the scaling window $p^{-l} \ll \ell_W \ll L$ only if the Ginzburg number $\mathrm{Gi} = g^2 L^{2\sigma-N}/\kappa_W^2$ is of order one or larger; otherwise finite-size rounding sets in first. Third, the factor $\Phi(s)$ makes the criterion periodic in $\ln\mu$ with period $\sigma \ln p$: the hierarchy is invariant only under the discrete dilations $x \mapsto p^{-1}x$, $t \mapsto p^\sigma t$, and the theory inherits this discrete scale invariance. A Euclidean theory with a continuous scaling symmetry cannot produce such log-periodic corrections. Their size is, however, controlled by the Poisson summation of Eq. (10): the first harmonic has amplitude of order $|\omega_1|e^{-\pi|\omega_1|}$, with $\omega_1 = 2\pi/(\sigma \ln p)$. For $N = \sigma = 1$ the relative amplitude is about 5×10^{-11} for $p = 2$, 2×10^{-3} for $p = 7$, 0.07 for $p = 31$ and 0.24 for $p = 101$. Log-periodicity is therefore a qualitative signature of a discrete hierarchy, measurable only when $\sigma \ln p$ is large.

Simulations. To test the prediction beyond power counting, we simulated the effective spike model directly on the tree G_l with $p = 2$, $N = 1$ and $l = 40$, with $w_{ij} \propto 2^{-k(1+\sigma)}$ for two neurons whose smallest common module has 2^k neurons [i.e., $\|i-j\|_p = 2^{k-l}$; this is Eq. (3) with $N = 1$]. The simulations follow standard approaches of nonequilibrium statistical physics; the procedure is described in Methods.

Table I and Fig. 3 show three results. (i) The controls are reproduced: the chain gives the one-dimensional DP exponents, and the branching process gives $\delta \simeq 1$, $\vartheta \simeq 0$. In both cases the fitted τ_S exceeds the exact value by about 0.03, which we take as the systematic error of the fit. (ii) For $\sigma = 0.25 < N/2$ the exponents are those of the branching process, $\tau_S \simeq 3/2$, and at $\sigma = N/2 = 0.5$ they are close to it, with deviations of the size expected from logarithmic corrections at $N = N_c$. (iii) For $\sigma > N/2$, τ_S decreases continuously, to about 1.2 at $\sigma = 1.5$,

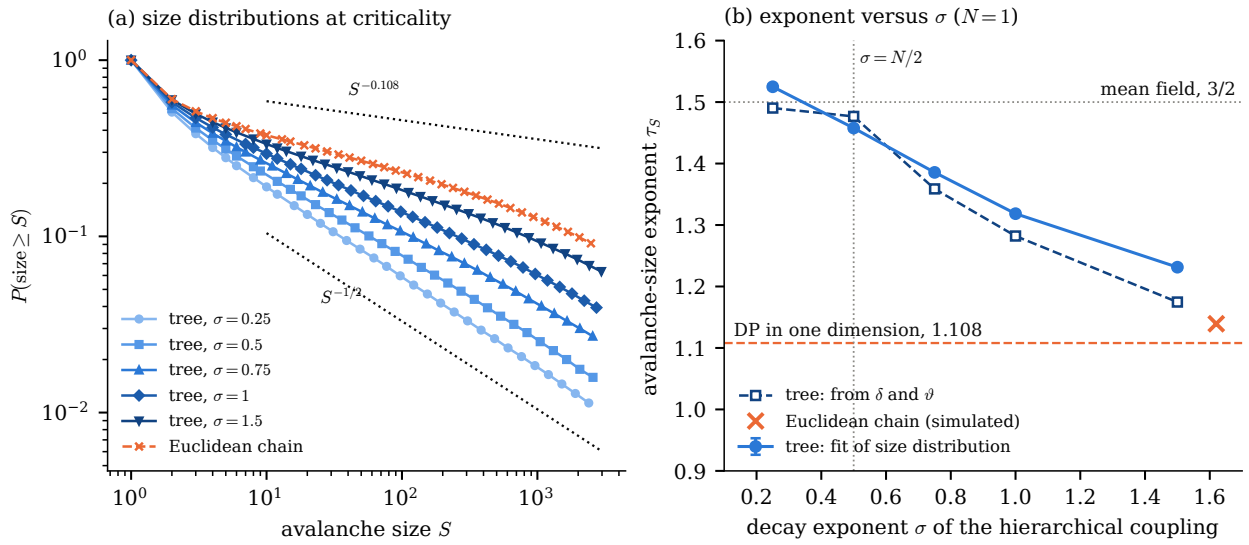


FIG. 3. Avalanches at the critical point. (a) Probability that an avalanche has size at least S, for the hierarchical network ($p = 2$, $N = 1$, $l = 40$) with five values of σ, and for a Euclidean chain with nearest-neighbor coupling. Dotted lines: the mean-field law $S^{-1/2}$ ($\tau_S = 3/2$) and the DP law $S^{-0.108}$ in one dimension. (b) τ_S against σ. Filled circles: maximum-likelihood fit of the size distribution ($10 \leq S \leq 2000$). Open squares: $\tau_S = 1 + \delta/(1 + \vartheta + \delta)$ from the measured survival and growth exponents. The vertical line marks $\sigma = N/2$, where $N = N_c$.

TABLE I. Critical point b_c (mean-field value 1) and exponents: survival $P(t) \propto t^{-\delta}$, growth $\mathcal{N}(t) \propto t^{\vartheta}$, and τ_S from the fit and from scaling. In parentheses, the exact values of $(\delta, \vartheta, \tau_S)$ for the two controls: the critical branching process, and DP in one dimension. The controls show a fit bias of about +0.03.

	b_c	δ	ϑ	τ_S^{fit}	τ_S^{scal}
Branching (1, 0, 3/2)	1.000	1.02	−0.06	1.53	1.52
$\sigma = 0.25$	1.106	0.99	+0.03	1.52	1.49
$\sigma = 0.5$	1.371	0.90	−0.01	1.46	1.48
$\sigma = 0.75$	1.739	0.65	+0.16	1.39	1.36
$\sigma = 1$	2.169	0.49	+0.24	1.32	1.28
$\sigma = 1.5$	3.157	0.28	+0.32	1.23	1.17
Chain (0.159, 0.314, 1.108)	4.848	0.16	+0.31	1.14	1.11

approaching the Euclidean value 1.11. The critical point also moves away from its mean-field value $b_c = 1$ as σ grows.

The mechanism is visible in the dynamics. On the chain a spike excites only its nearest neighbors, so activity spreads as a compact patch, new spikes keep landing on neurons that already receive input, and the saturation of f penalizes this crowding. On the tree with small σ the coupling decays slowly with hierarchical distance, most new spikes land far from the active neurons, crowding is rare, and the avalanches are those of a branching process. As σ grows, activity stays inside small modules and crowding returns.

Discussion. In the Euclidean theory with short-range coupling, the dimension of space decides whether neural avalanches are mean-field, and the Beggs–Plenz exponent requires $N \geq 4$. In the hierarchical theory the decay of the coupling up the hierarchy decides: $\tau_S = 3/2$ for $\sigma \leq N/2$, and non-mean-field exponents for $\sigma > N/2$. This yields a falsifiable prediction. Measure how connection strength decays with hierarchical distance, for example from connectome data organized by modules, and extract σ from Eq. (3). The theory then predicts whether avalanche statistics are mean-field or anomalous. A cortical sheet with $N = 2$ needs only $\sigma \leq 1$.

The theory applies equally to artificial networks whose dynamics is a hierarchical Wilson–Cowan equation, such as p-adic Wilson–Cowan and cellular neural networks [20, 21].

Several questions remain open. The critical exponents for $N < 2\sigma$ require the p-adic renormalization group [22, 23]; the simulations show that they interpolate between the mean-field and the Euclidean DP values, but their values are not yet known analytically. The simulations cover one tree ($p = 2$, $N = 1$) and one saturation scale. Finally, hierarchical networks with quenched heterogeneity show extended critical (Griffiths) regions [24]; the translation-invariant p-adic model has a sharp transition, and level-dependent disorder would interpolate between the two pictures. The companion paper [9] gives the full construction, the loop-corrected hierarchical Wilson–Cowan equations, the small-world property of these networks, and the phase transition and self-organization.

Methods. The simulations use standard methods for absorbing-state transitions [25]. *Dynamics.* The network has 2^{40} neurons, so that the boundary is essentially never reached; there is no self-coupling, the total coupling is $w_0 = 1$, $\alpha = 1$, and $f(s) = f_{\max}(1 - e^{-s/s_0})$ with $s_0 = 0.25$. The continuous-time dynamics is simulated exactly, by an event-driven (Gillespie-type) algorithm [26] combined with thinning [27]: since $f(s) \leq f'(0)s$, each spike attempts to trigger a spike at rate $b = f'(0)w_0/\alpha$ at a neuron i chosen with probability w_{ij}/w_0, and the attempt is accepted with probability $f(s_i)/[f'(0)s_i]$, so that the saturation is treated without approximation. *Spreading from a seed.* Each run starts from a single spike, as in the spreading method of Grassberger and de la Torre [28]. For each σ the critical value b_c was located by bisection, requiring the mean number of spikes $\mathcal{N}(t)$ to be a pure power law up to $t = 10^3$. At b_c we generated 2×10^5 runs and measured the survival probability $P(t) \propto t^{-\delta}$ and $\mathcal{N}(t) \propto t^{\vartheta}$ by least-squares fits on logarithmic scales for $30 \leq t \leq 10^3$. *Avalanche sizes.* The size S is the total number of spikes. The exponent τ_S was obtained by maximum likelihood for a power law truncated to $10 \leq S \leq 2000$ [29], with bootstrap errors, and independently from the scaling relation $\tau_S = 1 + \delta/(1 + \vartheta + \delta)$ [3]. *Controls.* The same program was run on two cases with known exponents, a pure branching process and a Euclidean chain with nearest-neighbor coupling (one-dimensional DP); they calibrate the systematic bias of the fit. The simulation code, the data and the scripts that produce the figures are available at `https://wazuniga.github.io/padic-BC/`. We used Claude (Anthropic; model claude-opus-5-5) to help implement the numerical simulation code for some well-known techniques in statistical physics Refs. [25–28]. The control was carried out as discussed before.

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[1] J. M. Beggs and D. Plenz, J. Neurosci. **23**, 11167 (2003).
[2] M. A. Buice and J. D. Cowan, Phys. Rev. E **75**, 051919 (2007).
[3] M. A. Muñoz, R. Dickman, A. Vespignani, and S. Zapperi, Phys. Rev. E **59**, 6175 (1999).
[4] O. Sporns, BioSystems **85**, 55 (2006).
[5] L. da F. Costa and O. Sporns, Eur. Phys. J. B **48**, 555 (2005).
[6] O. Sporns, G. Tononi, and R. Kötter, PLoS Comput. Biol. **1**, e42 (2005).
[7] E. Bullmore and O. Sporns, Nat. Rev. Neurosci. **10**, 186 (2009).
[8] E. Bullmore and O. Sporns, Nat. Rev. Neurosci. **13**, 336 (2012).
[9] W. A. Zúñiga-Galindo, p-Adic statistical field theory of fluctuation effects in hierarchical neural networks. In preparation; available at arXiv.org.
[10] M. Mézard, G. Parisi, N. Sourlas, G. Toulouse, and M. Virasoro, J. Phys. (Paris) **45**, 843 (1984).
[11] G. Parisi and N. Sourlas, Eur. Phys. J. B **14**, 535 (2000).
[12] V. A. Avetisov, A. H. Bikulov, S. V. Kozyrev, and V. A. Osipov, J. Phys. A **35**, 177 (2002).
[13] V. S. Vladimirov, I. V. Volovich, and E. I. Zelenov, *p-Adic Analysis and Mathematical Physics* (World Scientific, Singapore, 1994).
[14] W. A. Zúñiga-Galindo, *Pseudodifferential Equations over Non-Archimedean Spaces*, Lecture Notes in Mathematics Vol. 2174 (Springer, Cham, 2016).
[15] A. Bendikov, A. Grigor'yan, Ch. Pittet, and W. Woess, Russ. Math. Surveys **69**, 589 (2014) [arXiv:1304.6271].
[16] H. K. Janssen, K. Oerding, F. van Wijland, and H. J. Hilhorst, Eur. Phys. J. B **7**, 137 (1999).
[17] H. Hinrichsen and M. Howard, Eur. Phys. J. B **7**, 635 (1999).
[18] F. J. Dyson, Commun. Math. Phys. **12**, 91 (1969).
[19] P. M. Bleher and Ya. G. Sinai, Commun. Math. Phys. **33**, 23 (1973).
[20] W. A. Zúñiga-Galindo and B. A. Zambrano-Luna, Entropy **25**, 949 (2023).
[21] B. A. Zambrano-Luna and W. A. Zúñiga-Galindo, p-adic cellular neural networks, J. Nonlinear Math. Phys. **30**, 34–70 (2023), doi:10.1007/s44198-022-00071-8.
[22] É. Yu. Lerner and M. D. Missarov, Theor. Math. Phys. **78**, 177 (1989).
[23] A. Abdesselam, A. Chandra, and G. Guadagni, Rigorous quantum field theory functional integrals over the p-adics I: anomalous dimensions, arXiv:1302.5971.
[24] P. Moretti and M. A. Muñoz, Nat. Commun. **4**, 2521 (2013).
[25] H. Hinrichsen, Adv. Phys. **49**, 815 (2000).
[26] D. T. Gillespie, J. Comput. Phys. **22**, 403 (1976).
[27] P. A. W. Lewis and G. S. Shedler, Nav. Res. Logist. Q. **26**, 403 (1979).
[28] P. Grassberger and A. de la Torre, Ann. Phys. (N.Y.) **122**, 373 (1979).
[29] A. Clauset, C. R. Shalizi, and M. E. J. Newman, SIAM Rev. **51**, 661 (2009).